

CHAPTER 7

Blockchain for digital rights management

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1. Introduction

The emergence of the internet has brought in a huge number of resources in the form of information and sources of entertainment. Apart from being a hub for social media and communication gateway, it also acts as a source of virtual technology, giving a better experience to the users all around. Even though it acts as a revolution in its own way, it possesses certain disadvantages. The availability of millions of resources on the internet gives users a way to copy and share content as well as falsely providing ownership for the same. This increases the chances of piracy and a waste of talent and recognition to the real owners. Apart from massive availability of resources, it is equally important to add a layer of protection over the content available, so the internet becomes a safe place to share authentic content and provide desirable recognition.

One of the most important ways to curb the issue of piracy on the internet is digital rights management (DRM). DRM is a process of combining different secure technologies to provide protection to the owners and secure their ownership rights. It provides certain rules and guidelines across all forms of digitized content, mainly regarding the process of distribution and sharing. The data are converted into different formats and are flagged in such a way that they are available only to trustworthy devices and users. This prevents unauthorized users from being able to access content and illegally share precious digital content. These guidelines can also restrict useful processes such as backup maintenance, accessing the content of different devices by the same owner, and so on.

DRM is the technology that inhibits piracy and unauthorized use of right-protected material. DRM is applied on all levels of software and hardware from company logos to artistic works, digital sound, images, etc. (Ma et al., 2018a,b).

In recent times, digital rights management systems (DRMS) have become more useful and proven themselves time and again effective in various fields of digital technology production, development, and distribution. When the first DRM was discussed in the early 21st century, there was not clear guidance on what is defined as digital content and the law related with it, but as we now use more and more proprietary technology, the laws have become stringent, leading to a clear set of standard guiding procedures.

At first DRMSs were being applied only to research papers, blueprints, and information systems, but now, DRMSs are being employed beyond these industries.

Nowadays, DRMSs could be applied to most of the hardware components as the designs are made using digital technology. In the case of agricultural technology, John Deere used the Digital Millennium Copyright Act (Congress, 1998) to prevent local mechanics from recreating their fairly simple machinery parts. The world of DRM goes hand in hand with the intellectual property rights and finance sector. DRM management, often not discussed elaborately, is really important in the case of research and development, especially in the fields of science, engineering, technology, and medicine.

DRMS is one of the fastest growing technologies employed. The recent development of peer-to-peer (P2P) file sharing systems based on IPFS and blockchain technology (Vimal and Srivatsa, 2019), such as Torrents sites piracy, has become a huge aggravation for digital content creators. The digital content creation industry is expected to have market cap of USD 397,390 million by 2026 (Swatman et al., 2006). This figure does not include the mainstream digital content industries such as entertainment, e-books, patents, etc., but by including, the monetary value involved would reach a huge figure, proving that there should be a much more secure way of handling DRM.

In the COVID-19 scenario, as more and more businesses are being carried out over the internet, there is more possibility of internet theft. This is where we should hugely rely on the technology of blockchain.

Having discussed the intricacies of DRM systems from the origin of such systems to current applications, we can clearly make out that as technology of both content creation and digital piracy moves forward, we

need a more trusted model of rights verification that is able to build up flexible storage to meet enormous demand for content around the world, and obviously has a robust DRM protection scheme for fast identity verification and a conditional useful feature for tracing violations back to the original source of the content leak. Apart from these very apparent benefits of said P2P DRM system, another benefit will come to the customer's side. They will have an indefinite access to content that they purchase even after the content serving organization has been shut down.

2. Illustrations

To make DRM safe and efficient, it is very important to integrate the same with blockchain technology. This new process of using DRM with blockchain is also being introduced by many companies to protect copyrighted content. As bad as videos of unlawful distribution of rights in the blockchain and the loss of record-keeping rights have gotten, some new concepts have been introduced using the DRM blockchain.

A visual distribution system is introduced such as a blockchain and a functional version. A distributed blockchain-based license was also implemented and designed, but there was a flaw that the platform should have high performance to manage the top acquisition keys simultaneously.

The blockchain-based DRMS was proposed, but the design was not holonomic because only transaction details were recorded in the blockchain. After DRM analysis and DRM requirements came the DRM blockchain-based system, which supports ownership and privacy protection and conditional management compliance. The state-of-the-art machine is used to build a naming and storage system and to install crowds based on blocks where it helps to build a blockchain-based DRM platform.

2.1 Cryptocurrencies

Blockchain science can go hand in hand with the introduction of digital currency, which can add the economic layer to the blockchain. This 'scientific science' can be used to make micropayments to publishers who do not feed their content. It can also bring rewards for scientific work, such as peer review, statistical support, laboratory exchanges, the release of specific research, or hosting of data. Finally, initial coin donations, a form of refund using cryptocurrencies, can be used to fund all research projects. In this way, the crypto economy can be transformed into a science that shows value suitability for multiple jobs.

2.2 Content distribution and DRM

The role of the blockchain has been largely researched (i.e., nonacademic), where online migration has led to a change in the distribution of revenue from content creators and publishers to hosting companies, communications giants, and advertising consultants. To some extent, this change is due to the natural feature of the World Wide Web, namely the use of hyperlinks.

Hyperlinks are one-way content links: they do not point back to users who click them. Therefore, there is no way to allow small automatic payments (micropayments) to be used. Given this, the only options for publishers are to either open content and support business modelling in advertising or to place unpaid payments on expensive credit card bills. Micropayments can also create another way for business models prominent in educational publishing (subscriptions and open access), each coming with its own challenges.

2.3 Storing research data

The low-blockchain environment paves the way for the creation of a database for research activities from all research environments that can be shared collaboratively. Although creating this database is technically possible with current database technology, there is a requirement from the central gatekeeper as well. The owner of such a data store makes its fulfilment less likely. This low-level data store will show that whenever a researcher, for example, uploads data, performs statistical analysis, writes and presents an article, or reviews a manuscript, this will be automatically tracked and recorded, researched, and very productive. The risk of fraud will also be reduced. In addition, it will be repeated. It is much easier to gather reliable and complete data on the performance of research, so researchers, as well as universities, can allow for the construction of higher and more reliable metrics more. In addition, it will allow metrics to be based on tasks that are not yet fully developed (e.g., peer review).

3. DRM requirement

Over the ages the evolution in techniques of producing content and distribution of the same in general has gradually become simpler for both consumer and producer. Earlier, to listen to music, one had to visit a music store nearby and purchase a cassette loaded with the songs or their choice

and had to purchase a cassette player to play the song. Today it takes a matter of seconds to download and listen to songs through a mobile phone and the wonderful possibilities of an internet connection. In the traditional case of DRM, it mainly focuses on preventing illegal usage of digital content without paying any monetary benefit to the concerned content creator.

There was a physical aspect that the content and rights of the content once sold did not provide the creator a lasting benefit but rather only a one-time payment, the rest going to the middlemen that distributed the content. Once the content gets encrypted, there is no possibility for auditing the content in it. In recent times, as distribution of content has crossed physical barriers, the need for verification of content and the credibility of the creator has tremendously increased. For example, to prevent underage kids from watching movies with violence, illegal, or sexual content in them, thereby enforcing proper certification of movies, verification of authenticity of online distribution sources is needed since the internet in all of its glory is a free space to trade and distribute such content and much more (Karuppiyah and Saravanan, 2014, 2015; Karuppiyah et al., 2016, 2017, 2019; Kumari et al., 2016; Naeem et al., 2020; Maria et al., 2021; Pradhan et al., 2018; Li et al., 2017). Many such scenarios give rise to a new set of security and requirements in DRM as follows:

- content verifiability along with tamper proofness
- privacy of identity for digital content provider
- protecting the access of content
- usage control of content
- trademarking and licensing
- detection of violation

3.1 Content verifiability along with tamper proofness

Content made available for public purchase has to go through some legal procedures and audit before being hosted on a public platform. It has to follow rules such as the platform specifying that the content be verifiable as well as auditable. It should also be made clear to the platform's owners and distributors what clear track was taken to acquire the content, as a sequence of how it exchanged hands or the chain of custody of the content being uploaded.

Once it has been verified that the content has been acquired from a legal source, it is then checked thoroughly for vulgarity in accordance to content

distribution acts. If the content is being reported as abuse or vulgar the hosting platform must be in position to hold the creator accountable for this, to prove that tamper-proof evidence is required. This eliminates any future need to detect and seal leaks of content as the content acquired is ultimately leak-proof since no party can exchange ownership by simply downloading and tampering with creator metadata.

3.2 Privacy of identity for digital content provider

The rights management system must be in a position to prevent an adversary from exploiting the privacy and access of the user. It must also be in position to hold a user accountable for his action like spreading abuses, rumors, violence, hate speech, etc. In general, there should not be any bias toward anyone in the system.

3.3 Protecting the access of content

Before the content is uploaded to hosting platforms, it must have some protection to prevent piracy and to prevent someone else claiming the credit for another's work. Usually, cryptographic encryption or watermarking is done to ensure this.

3.4 Usage control of content

As the access to content is prevented, it should include sophisticated management mechanisms for further usage of content and also to track its usage.

3.5 Trademarking and licensing

When a user uses the content available in the hosting platform, he/she agrees to the terms and conditions of the license under which the content is published. The license specifies the terms such as usage times, period, domain, rental, translation, or watermark that defines the ownership of the content.

3.6 Detection of violation

During the usage of the content, the user should always be bound by the terms and conditions specified by the license. The DRMS should be in a position to monitor and track when violation happens.

4. Parts of a traditional DRM

A traditional DRM system has multiple components that have immense functionalities that need to be properly understood. The main components of a traditional DRM system are as follows ([Fig. 7.1](#)).

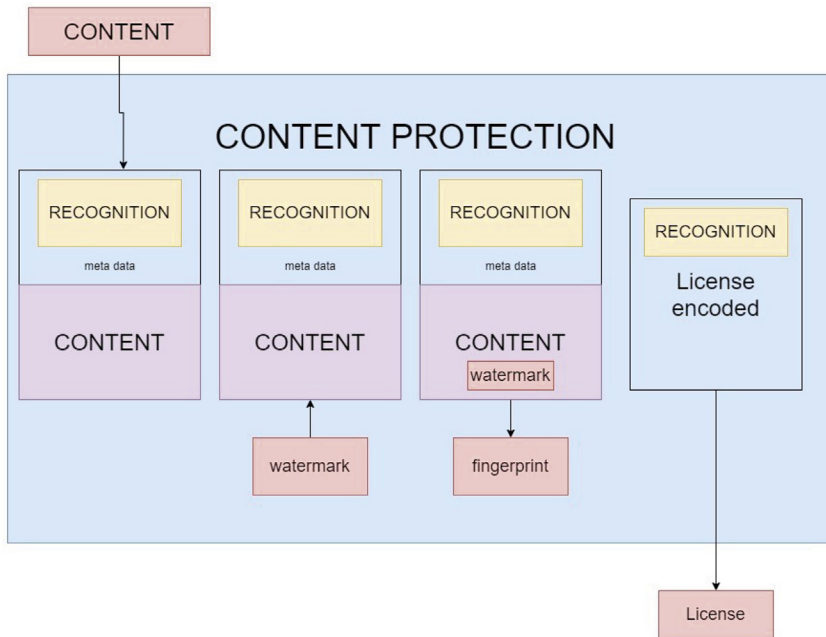


Figure 7.1 Components of a DRM.

4.1 Content recognizer

The main property of a content recognizer is to remain consistent. It is often said, even if the rights of the content change, the content recognizer should always remain consistent. To maintain consistency and generate persistent systems, the content recognizer uses standard identification schemes for numbering and copyrighting such as International Standard Book Number (ISBN), International Standard Serial Number, and Digital Object Identifier System, etc.

4.2 Meta data

The meta data include different schemes that are used also with the content recognizer to provide more insight about the content and how to access the information.

4.3 Watermark

A watermark is another important technology to prevent the unauthorized access and copying of data and for identifying and tracking data. Most watermarks use a traditional approach to use lower amplitude signals that

are inserted into the content directly. The watermark can be easily identified using different methods that can be correlated by authorized hardware and software stores.

This technology helps to easily detect copies of data being sold illegally that are not properly protected from such kinds of attacks. However, it is very important to ensure that the watermarks injected into the content do not hamper the content and are not easily identifiable by hackers and attackers. It is also important to detect and remove fake and falsely added watermarks from authentic content.

4.4 Fingerprints

Fingerprints are generally used to address data based on its content by matching it to components of data stores or databases. It is very different from watermarks in many ways:

- Watermarks get injected into the system, whereas fingerprints do not get injected, so they do not change any content.
- Watermarks require the availability of content to get injected; fingerprints do not need the content to be available.
- If any kind of changes in the content take place the watermarks need to be properly processed again over the data as well as all copies that are distributed across the internet, which is not at all required in fingerprints.
- Fingerprints need to be stored for any kind of new content that is to be added in the database, which is not required in watermarks.

4.5 Containers

Containers provide security for the content. They use different algorithms like hashing algorithms to secure the content. The digital certificates coupled along with these encryption algorithms also help in providing confidentiality for the data.

4.6 Security licenses

These are used by users specially to mention their authoritative laws and rights that guarantee integrity of the content. The security license is only available to a particular device and is not easily transferred to any other device.

4.7 EUP

End user player (EUP) is another component of the DRM. It acts as a way to make the rights of the user to the content digitally available across the

network. It is used to restrict any kind of actions to the content by the ones accessing the content. These components act as the backbone of the DRM technology and also help in providing security, integrity, confidentiality, and secure availability of the content across the internet. Even after so many layers of protection, DRM technology still has loopholes that make it desirable to use blockchain along with it to increase security of content.

5. Compatibility of blockchain for DRM

5.1 Immutability

Immutability is one of the most important properties of blockchain. It states that the blockchain technology is permanent and write-only. Every node maintains a copy of the ledger, so if some new content is being added or updated, every node has to check the validity of it and add it to the ledger. This makes sure that DRM is transparent and free of corruption. A public blockchain can be used in case of public DRM, whereas enterprises can use private or consortium blockchain for ensuring privacy from outsiders and maintaining its inner transparency.

5.2 Decentralized

A decentralized network means that it does not have any governing authority or single person on top; rather a group of nodes maintains the blockchain, making it decentralized. We can store content, as we have direct access over it with our private key. The decentralized network ensures that everyone has equal rights and power regarding their assets. This ensures that our chain experiences less failure, good user control, is less prone to break down, has no third-party, zero scams, and has a high level of transparency and authentic nature. This is key for a nonbiased DRM.

5.3 Enhanced security

As we have gotten rid of central authority, it just does not mean that anyone can modify the characteristics of DRM. Usage of cryptographic hash functions adds a layer of security. We know that hashing is irreversible, so if an adversary wants to destroy the DRM chain, he/she would have to modify all data stored on every block of every node in the network. There could be billions of people, where every individual has the same copy of the ledger. Updating the hash in millions of computers is next to impossible and costly.

5.4 Distributed ledger

Usually a distributed ledger will provide all the information about a transaction and its user, thereby ensuring fairness in the system. Because of this distributed power of computing across the computers a better outcome is ensured. This helps the DRM to achieve no malicious changes, managership, and quick response.

5.5 Consensus

Every DRM blockchain is powered because of the consensus algorithms. The architecture is essentially designed, and consensus algorithms are the core for this architecture. Consensus in blockchain makes it essential for the decision-making process. This takes care of choosing the next node and repairing its earlier mistakes. This consensus algorithm is responsible for maintaining the trustworthiness of the DRM chain.

5.6 Faster responses

Traditional hosting platforms are quite slow. Sometimes it can take days to process an upload and get benefits from it. It also can easily be corrupted. Blockchain provides a faster approach compared to traditional DRM systems. This way a user performs uploading relatively faster, which saves a lot of time in the long run.

With all the aforementioned advantages, blockchain is a perfect companion to DRMS. Generally the effect of blockchain impacts the way in which the digital world works to another level (Singh and Kumar Tripathi, 2019) (Fig. 7.2).

6. Various cryptographic hash functions in blockchain

6.1 MD5 algorithm

The MD5 algorithmic function, also called the MD5 message digest algorithm, was invented in the late 1990s by mathematician Ronal Rivest. It is the successor to the MD4 algorithm and proves to be much more effective as well as efficient. At first, it was only employed to be used as a crypto hash, but it was found to suffer from vulnerabilities so extensive that a relatively weak machine was able to crack it and produce collisions in polynomial time. However, now widely used as a checksum algorithm to prevent an advisory changing the content of the blockchain, it still is suitable for other noncryptographic applications and for operations with regard to partitioned databases.

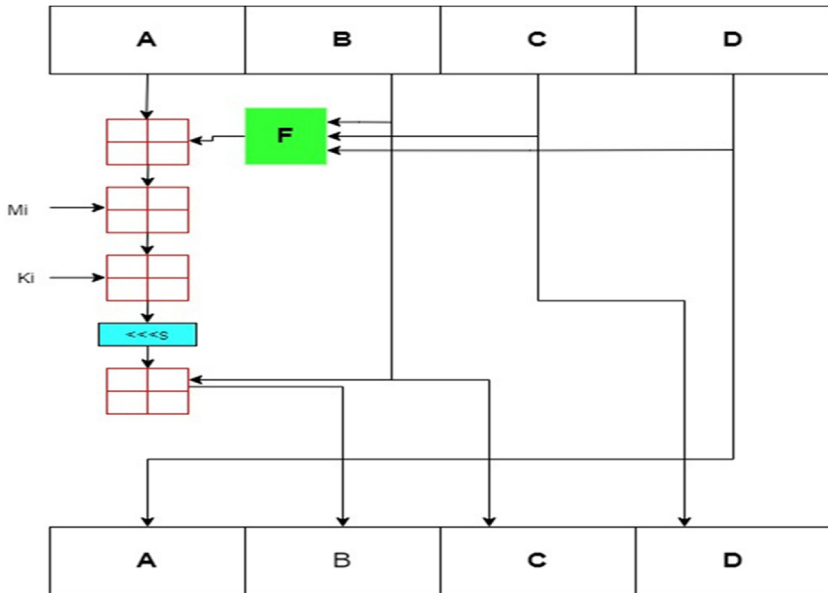


Figure 7.2 MD5 algorithm working.

Talking about suffering from vulnerabilities, what are these and how does MD5 perform in relation to them? Hash functions employed in cryptography are considered powerful based on many features, one of them being their property to not feasibly produce multiple messages with similar hash values.

A collision attack can usually find the solution in seconds of the MD5 hash. With advancement of the off-the-shelf GPU and TPU the complexity to find the collision is tremendously reduced. It took 218 operations to find the collision. It is still widely used for simple use cases even after the warnings by cybersecurity experts and is mostly depreciated.

6.2 SHA-0 algorithm

The SHA-0 hashing function produces a 160-bit hash value. The hash values are then given to a digital signature algorithm (DSA), which then authenticates or generates the signature for the hash values that are fed into it. It was first published as a message digest algorithm and was much more effective than its predecessor MD series.

A collision usually can be found in 261 operations; later improvised attacks found collisions in 236 operations. The US National Institute of

Standards and Technology stopped recommending SHA-0 from the year 1996. Although the use of SHA-0 is highly discouraged, researchers are still working on SHA-0 as it is predecessor of SHA-1.

6.3 SHA-1 algorithm

The SHA-1 hashing algorithm function generates a 160-bit hash value. It proved to be better than its predecessor SHA-0. It was designed by a US federal agency and then made open. Google published that it has made successful collision attacks on SHA-1. At last, it took 263 operations to find the collision. It took equivalent to the computing power of 6500 years on a CPU and 110 years on a single GPU computation. Theoretically speaking, SHattered attack is 100,000 times faster than the brute force attack used to perform a collision attack on the SHA-1 algorithm. The NSIT and other cybersecurity experts have depreciated its use on any collision avoidance requirements.

6.4 SHA-2 algorithm

The SHA-2 has similar hash functions with varied block sizes, such as SHA-256 with 32-byte words and SHA-512 with 64-byte words. It consists of SHA-244, SHA-256, SHA-384, SHA-512, SHA-512/224, and SHA-512/256. SHA-2 hash function is widely used in security applications and protocols including TLS, SSL, PGP, and IPsec. It uses from 64 to 80 rounds of cryptography operations, and it is commonly used to validate and sign digital security certificates and documents.

6.5 SHA-256 algorithm

This algorithm is used to get a constant 256 bit every time. It is the algorithm that is used in blockchain technology. The complexity of this algorithm is 2211.5. This hash function's collision resistance complexity is highly important for the security of this DRMS using blockchain (Fig 7.3).

7. Methodologies and technology in use

Some of the blockchain technologies for DRM are as follows.

The blockchain system is being used to manage and protect rights of videos and graphics, as stated by Kishigami in their paper (Watanabe et al., 2016). This helps users to manage copyrights and maintain permissions to use graphics and other content. To view the content, the client should not be offline and should also have a transaction ID to decrypt the content.

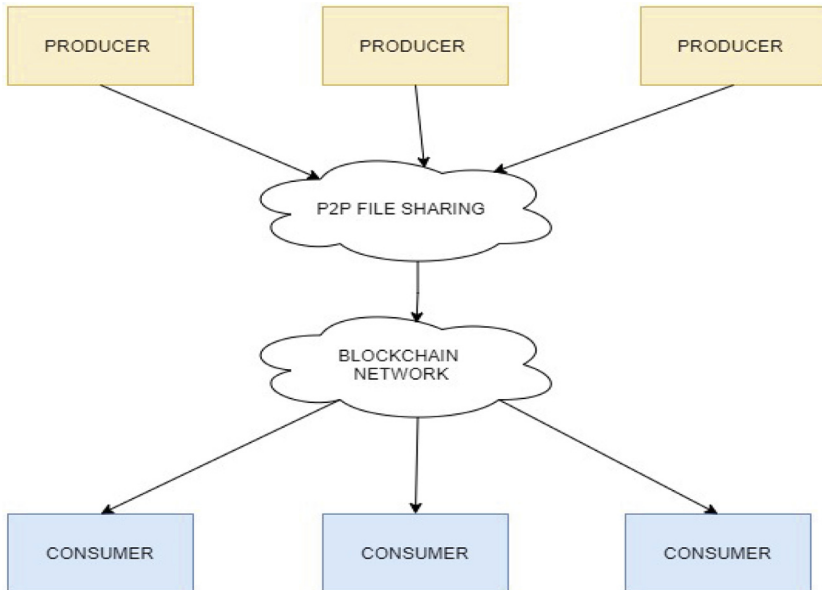


Figure 7.3 P2P file sharing architecture using blockchain technology.

Blockchain technology is also used to validate the license of software applications. There are mainly two methods of blockchain for validation processes. First is the master model of Bitcoin, helping the user to store more information like time limit of expiry, etc. The other model, called the Bespoke model, is used for one user having one license. This technology will be further used for holders having multiple licenses available.

The ownership of image records is also performed using blockchain technology. The records are stored in the form of timestamps and other services, which helps in storing the information of the image owner. A machine learning algorithm is used to find out images being used in different websites without the permission of the owner.

The Ethereum blockchain technology is used in copyrighting music content. Copyrighting involves protecting payments, music compositions, owner rights, distribution rights, and most importantly production costs. Peer music is another technology being used for music copyrighting for the peer tracks. The muse blockchain being used here provides cryptocurrency to different users on the basis of consumption by different content creators and music artists.

DECENT is a customized blockchain technology that uses delegative proof of stake for distributing content. It helps to create a system where content is rated and cannot be changed or modified. It builds custom tokens allowing the artists to create further methods for earning cryptocurrency.

8. Effects and applications of using blockchain in DRM

8.1 Effects

8.1.1 Discoverability of copyrights

Users can query for different copyrights and their history in blockchain. This can be done with the help of unique identifiers that will be used for identification. Collection of the copyright history will help to deduce a route of ownership rights.

8.1.2 Privacy

The specifics of production copyrights are only present securely with the owner. Cryptography technology states that the content present in them cannot be retrieved using a hash digest in a copyright record. Hence, the detailed content present in the design can be protected.

8.1.3 Efficient method

The authorization of the copyrights using the blockchain consensus algorithms is very efficient. In the Bitcoin network, where the processing speed is considered relatively low, it takes about a week to complete all the copyright registrations for the whole year. This is relatively lower in comparison to the present methods of copyright management.

8.1.4 Maintenance of security and fairness of copyright trading

If anyone tries to provide false information while registering for a copyright, fraudulent evidence left will definitely get caught by the blockchain technology. In this case, the buyer will get punished by authorities for the fraud records and information provided by them using their digital signatures. Fraudulent copyright records are not protected. The use of blockchain will hence guarantee the fairness and security in copyright management. It will also help in maintaining authenticity of copyrights to a certain extent.

8.1.5 Protection from repetitive registrations

If a user registers the same work multiple times, it will generate multiple IDs for the same work. Even though he will have multiple copies of his

work, the person will not be able to sell the work again and again, thus stealing the work recognition that has been sold once. The repeated trading of the same work, on the contrary, will leave digital evidence of piracy in the blockchain technology.

8.1.6 Default in reception

Copyrighted data can be obtained by the buyer only when he sends the private key and then receives its delivery and decrypts it. If the buyer has not requested the decryption and delivery, the actual information will never be retrieved through any other method. Therefore, the security of the information can be provided. A buyer sets a contract to set a deadline to sign for delivery and decrypt it. If the deadline is not met, the default process will get started.

8.2 Applications

8.2.1 Educational learning passports

Today, with the development and increase in the number of sources available to learn, it is difficult to maintain records of all the lifelong learnings and achievements. Blockchain technology with DRM integration can be used to permanently store achievements, certifications, and records digitally in the form of educational passports, which will act as a proof of educational experience for individuals. The advent of educational passports with unique identification factors will help increase the value of certifications and additional educational skills that can be acquired by an individual (Guo et al., 2019).

8.2.2 Multimedia community of educational resources

Blockchain technology can also be used as a measure to maintain rights and security of educational resources, thus creating a secure community of educational resources by authorized and experienced people in the industry. Learners will also be able to securely share their learning resources through the multimedia community as educational creators, which will further help to increase availability of educational resources (Van Rossum, 2018).

8.2.3 Copyrighting multimedia works

The application of blockchain technology can also be extended to apply to protection and digital rights maintenance of multimedia works, which will help in developing a web of diverse multimedia resources and their digital rights maintenance. Law and technology have a tedious relationship. In

recent times, with the boom of development in technology, development in legal fields is being accelerated, while the technology is also being shaped because of law. The development of law for copyrights proves these efforts as its enforcement and development are usually closely inspired by the technologic state of the art. With the emergence and ease of use of blockchain technology and blockchain-based contracts, some people perceive a recent wave of technical transmutation that could have a long-standing impact on copyright law. Indeed, it is recommended that these technologies help us to provide new technologic mechanisms for the sole purpose of managing the intellectual property rights that can now be implemented in current scenarios with a wave of development in over-the-top media services (Cho and Jeong, 2019).

8.2.4 DRM in the music industry

Blockchain technology can offer solutions to problems inside the music industry that are often reported by artists, composers, and producers. The technology may not be able to help solve all the problems, but it may show a way out of the current problem between artists and middlemen by getting rid of the outdated and old procedure. This will help an artist to show their original content without any hesitation, enabling a more secure way of licensing their work and having their copyrights properly secured.

Blockchain may help in transforming the music industry, such as challenging the economic models and bringing about a more equal place for all artists, not just record labels and production houses.

8.2.5 Blockchain and copyright law

At its core, blockchain is in fact a shared and in sync database that is supported by an algorithm, and there are multiple nodes involved in this ecosystem, where each node contains an exact copy of the complete database. The blockchain achieves a high level of resilience since every node has a copy of the blockchain, whereby even if one node gets corrupted, we can be sure that the chain remains intact. In such cases the possibility of a single point of failure is avoided. The technology derives its name from the fact that it is ordered as a chain of individual blocks. Every block comprises the transactions that happen during the digital media being procured, maintained, and delivered. Every transaction is transformed into a block as it reaches a certain size; this gets appended to the existing ledger by a process of hashing. Not just any node can append their block; as

mentioned earlier, a node proposes a block, which is displayed to every other block in the ecosystem. Then a node assumes a role of miner, which plays a key role in hash generation (Savelyev, 2018).

The blocks contain different important components in them including a timestamp, a hash of the previous block (which creates a chain of sequential blocks of transaction in chronological order), and the hash of all transactions contained in the block (its “fingerprint”). Since only data can be added to the blockchain, we can safely say that it is only an appending data structure.

There are actually quite a few different flavors of blockchain based on their technicality and governing constraints: “private and permission” and “public and nonpermission.” In a public chain, any individual can download the ledger and start using it. The transaction is open to the public. This provides more transparency but also comes with a disadvantage of less privacy. In the case of private chains, it is the opposite: there is an authentication process before anyone can use the chain, which provides more privacy. The private chain is sometimes called an enterprise chain. In the case of using blockchain in DRM, a combination of both would be fruitful (O’Dwyer, 2020).

Simply, a narrative smart contract is a set of self-executing programs that process input as it gets triggered. It is a small set of computer scripts that are deployed in blockchain. As smart contracts are currently being discussed alongside blockchain, similar techniques have been used by DRM for a long time. As we mentioned earlier, DRM implants copyright law into the digital content by suppressing the user’s ability to copy, view, print, play, or other works. For example, audio files embedded with DRM technology are not subjected to the double spending attack because they consist of a basic smart contract, which references a centralized web network (for example, Spotify’s server programmed to enforce the terms and conditions of the application).

The blockchain technology in copyright law mainly works around the following basic logic: 1. There is the ability to identify digital content and monitor it, thereby protecting the fluidity of the document. 2. The idea is to promote transparency and disintermediate the occurrence of transactions. 3. There is the ability of blockchain to be adapted as a DRMS.

The early origins of blockchain and DRM are quite distinct. The first was developed as a P2P network for the Bitcoin community. Nevertheless, as tension arose between DRM and copyright law, a similar case is likely bound to happen between blockchain and blockchain contracts. This is an

important point that we have to remember. Distributed ledger technology (DLT) is still immature and can be easily molded. The further development also lies in configuration of DLT chosen for implementation. There is also a minor danger that we might lose the public policy aspect of blockchain, like DRM, which must be taken into mind during development (Finck and Moscon, 2019).

9. Methodologies for coupling DRM with blockchain

9.1 Blockchain implementation in the design industry (Lu et al., 2019)

Design works of the digital age need to be protected by digital technology, but this proposition fails ahead of advanced copying and piracy techniques, posing challenges for protection of copyrights.

Digital watermarking of content as earlier discussed coupled with encryption of content treated as a whole is common among DRM systems for protection and verification. Using these methods not only provides ease and moderate security, but it is quick and can be built into the content distribution software with no separate infrastructure.

Apart from the security concern of the usage of this method, watermarking and encryption of software pose other risks such as no protection against unintentional leakage or difficulty in presentation to potential buyers. These problems are the most common in the design industry (Lu et al., 2019).

Proposal for a blockchain-based system to be coupled with DRM is easy in theory as all there is to it is a blockchain with rights information as well as a record of transactions. The actual system though is not as simple to implement. Every other type of digital content like photographs, music, video, etc., will require a complete overhaul of the system.

Talking about methodologies adopted for a blockchain-based DRM system, there has been significant work in the past. DRMChain (Lu et al., 2019), being the most noteworthy, utilizes two separate BAIs or blockchain application interfaces, similar to application program interfaces, that help applications to interact with underlying hardware through a series of calls as instructions to store relevant metadata of the digital content.

DRMChain makes smart use of the BAIs, one to store the direct summary information of the data and the other to store the summary of the cipher produced for the data. As expected, digital content can be huge, so a

flexible external storage space for the original and cipher content is also proposed with hash IDs of content automatically linked to the relevant block in the blockchain.

DRMChain comes out to be a very secure and fast system with verifiability and usage control as main priorities. It applies multi-signature-based traceability approaches, ensuring efficiency in retrieval of constraint and consumer transaction lists from the blockchain to gather free or paid consumption record history.

9.2 Scalability in blockchains (Garba et al., 2020)

Another popular model of a blockchain-based DRM system utilizing overlay networks to enhance the scalability and speed of the system has been proposed (Garba et al., 2020).

Decentralization, security, and lastly, scalability have been identified as key essential features of blockchain technology, but achievement of only two out of these three crucial features in modern day blockchain applications is seen. Scalability is clearly neglected in today's first-generation blockchain technology, which suffers from immeasurable transaction costs to high energy consumption both in the computation and power contexts.

High latency with visibly slow throughput also riddles these applications. Overlay networks are used in this type of implementation that are different from the legacy implementations using small, centralized networks to transport blocks over the whole system. An overlay network implements cluster heads that blindly forward the block that is given to them by cluster nodes. There are multiple clusters with innumerable number of nodes each with a cluster head.

Nodes in a cluster can transport their blocks to the cluster heads or to their neighboring nodes and then send them indirectly to the cluster heads, ensuring no discrimination by the cluster heads in transporting blocks. The blocks are encrypted and the key to encryption is only made visible to the network once the block has been received successfully at the destination node.

9.3 Proof of stake consensus methodology (Watanabe et al., 2016)

As the previous methodology discussed some of the scalability issues in blockchain technology, the heavy use of computational power to validate and attach each block to the chain, also known as proof of work, can also prove to be a limitation for some applications of this technology. Proof of

work mechanism also wastes a lot of electricity toward unfruitful causes, resulting in the proposal of a newer kind of proof of stake (Watanabe et al., 2016) mechanism for validation of blocks.

A block is mined with a probability proportional to the amount of effort done by a miner under the proof of work validation system. Meanwhile, for a proof of stake type of validation mechanism, to validate a block, the miner has to provide proof of the amount of coins they hold and in which blocks. Once verified by all peers, the block is mined, and no more computational load is to be resolved.

Such a system acts for the removal of two drawbacks of the system. First, it solves the problem of a blockchain implementation of any application being very power hungry. Secondly, it also helps prevent monopolization of the workload by a single peer with the most powerful mining rig. If a peer has the strongest system or the ability to solve the puzzles the fastest, they will be the one getting all the mining tasks, establishing a monopoly over the blockchain.

With a proof of stake mechanism though, a peer in the blockchain will have to hold more than 50% of all the coins in the blockchain to be able to monopolize anyone, and that is already a very high mountain to climb. Also a deterrence mechanism is set up where the value of the coins held by the monopolizer will be significantly reduced, so the attack will be stopped.

9.4 Master-slave blockchain architecture (Ma et al., 2018a,b) implementation for diversity in mining rigs

Since the genesis of P2P decentralized currency platforms such as Bitcoin, a common question has been raised regarding the lack of ability of anybody on the planet to participate with their own computing device in the heavy lifting and mining of the coins without a dedicated mining rig. This is also recognized as one of the major flaws of the system. A master-slave blockchain architecture or MSB is proposed (Ma et al., 2018a,b). Such a system is recognized to be suitable for mobile devices, home PCs, and tablets for pervasive computing.

An already decentralized system such as blockchain is proposed to be further split into two layers. A master node is established and tasked with collection of transactions and distribution of computing tasks to the slave agents. Slave agents are established in a star-like topology with their own master nodes and are tasked to compute and commit computational exercises that they are assigned to by their masters. A low-power device such

as a personal desktop computer or mobile device such as smartphone or tablet may be the slave machine and a high-power server such as a supercomputer can be the master.

Whenever such a system with a central authority that assigns tasks to dedicated individuals in a network is proposed, a simultaneous system to efficiently assign the said tasks is brought under question. Two approaches, namely, noncomputing model (NCM) and partially computing model, are proposed under the same MSB blockchain architecture. NCM uses equivalent proof of work (PeW) consensus, while proof of contribution is used for partially computing model.

When the master node has dispatched computing tasks partially to each slave, the nonce value found by each slave is verified, and the slave producing the correct nonce is rewarded. For the NCM model, only the master does the heavy computing tasks, as usual, while the slaves have to provide service-related contributions such as visit view and unique visitor, etc., and are given a part of the reward that the master gains according to their contribution in the service-related sector.

9.5 DRM approach based on blockchain with automatic license assignment

Getting back to preexisting DRM technologies implemented using blockchain, Zhang ([Zhang and Zhao, 2018](#)) presents an implementation that records license information and copyright transactions, so all records are safer and more transparent. Reliability and integrity of copyright transaction issued is ensured using smart contract and assignment of licenses automatically, thereby eliminating the need for centralized servers.

Of course, the system has the functionality to provide copyright owners the ability to set prices to content as well as the mode and channel of content delivery. An example of such could be seen in streaming services, such as Netflix, that allow, say, content viewing up to 1080p quality and on a single screen at a time for its base subscription pack.

These are known as content usage rules. Customers are also empowered with their own usage rules and flexibility in payment frequency. As with every new technology that solves issues of the past and is commissioned for public consumption, blockchain-powered DRM systems have to abide by standards set in place to ensure fair use of digital content, and the proposed implementation presents an easily promotable license structure designed by closely keeping in mind current DRM standards.

Copyright transactions are recorded in blocks for reliability and transparency. Smart contracts are used for lasting reliability of each transaction. Overall, a credible and authoritarian system for content producers at the individual as well as upper levels is provided that ensures scalability and safety.

We see the proposal of a new consensus mechanism called Tendermint (Zhang and Yin, 2019, Buchman, 2016), which is a consensus engine for blockchain for simplifying implementation of the blockchain.

10. Advantages of integrating blockchain with digital content

10.1 DRM approach based on blockchain with automatic license assignment

There will always remain a demand for digital content, and people will continue to access and use it. New digital media content like songs, movies, videos, compositions, news articles, and photos are the sources of entertainment for the mass users. As the content in the digital media increases, it becomes much more difficult to handle the millions of transactions that take place every second for the consumption of such content. The illegal copying of data causes the increase in the transaction costs and deliberately leads to an inefficient system of distribution.

As the advent of blockchain makes the copyrighting system more efficient, it is hard for people to copy resources and send them illegally, which provides more profit to the content owners and the content creators, thus leading to a change in the pricing schemes for the same. This eventually leads to lost-cost settlement between content owners and buyers. Currencies involving Bitcoin technology like Bitcoin and Ethereum also provide a feature to undergo smaller transactions such as transaction cost of listening to a single track music, watching a single video, or reading a news headline. It can also help in providing more monetized options, which require fewer or no ads. All of this uses blockchain to help in fluid transactions of the content available across the internet and more options for the same.

10.2 Blockchain helps to bypass the middlemen present between the transactions

In a traditional transaction system, where transactions for content take place between creators or producers and the users, there is sometimes an involvement of a middle person or agent, which increases the overall cost of accessing the content when data finally reaches the user.

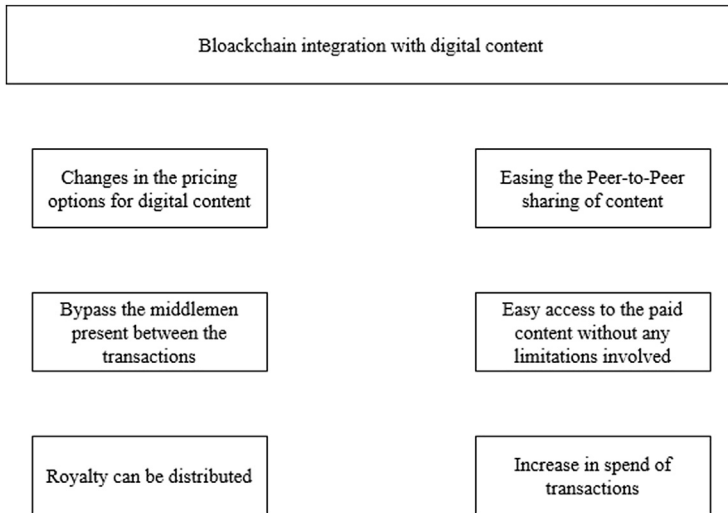


Figure 7.4 Blockchain integration with digital content.

Blockchain technology helps all users present in the market, either the sellers or the buyers, to become their own distributors, as the records of the transactions can be easily tracked and maintained. Nowadays, each person can create content from their homes and can use it to perform transactions. They can also list their sources of advertisement and can directly get involved with the end users without the need of an advertiser or a middle agent with the help of blockchain technology. This will also help in the transparent and proper tracking of revenues, as the artists can generate content on their own and, by using blockchain, can track its distribution across the globe. This will help make the market involving advertisements more liberal and provide options for better opportunities to track advertisement processes (Käll, 2018) (Fig. 7.4).

10.3 Payments including royalty can be easily distributed

Since blockchain eases the way out for content creators to showcase their content, different productions and companies provide contracts to the creators and pay them some fee in return, called a royalty, if they use or show their content in any way. Some of the examples include music productions, show music of independent artists, film productions using video contents in their films, and especially television channels showing advertisements to generate revenue.

However, there are many problems and complexities that can occur while settling the royalty payments. The actual amount depends on the number of times the content is shown by the company or agency, which is not linked in any way with the creator. This leads to generation of royalties that are vaguely generated and are not fully correct in reality. Thus, artists and creators do not get their desirable share of the royalty payments, which acts as a disadvantage.

The usage and distribution of content is controlled by the blockchain technology, also helping with precise and proper analysis of payments with the royalties. This helps in reallocation of these payments. A blockchain system can track the actual usage of the content being used by a company after being associated with the content owner. Credits are accumulated and stored in this system, and the content owners are then paid according to the share they deserve.

10.4 Easing the peer-to-peer sharing of content

The traditional P2P networks that have been used for a long time still possess many flaws and disadvantages. Since the content involved is so vast, it is very difficult to maintain a record of the data that is exchanged from one to another network.

The P2P content usage and sharing can be made transparent using the blockchain system. It helps in digitizing the transactions by producing bills that are automatically generated and easy to maintain. The billing systems are purely automated and are also fully controllable.

This provides a better way of regulating content for which money has to be paid by the users, and it provides an additional way of generating revenue for creators and right owners. This can also be applied to data copy that are shared to the end users, when the physical contents are provided on a blockchain (Fei, 2019).

10.5 Easy access to the paid content without any limitations involved

All the boundary limitations that are set by states or countries to the content providers are limited by the use of blockchain. The DRM complexities of the system are also decreased by the use of blockchain. Blockchain authentication helps in linking the contents that are directly consumed by the individual users.

The blockchain system also has the ability to retire the DRMS and decrease its availability, as all the transactions that are made available are

always tracked by the blockchain and fully linked to the owner and copyright holder. The money is generated automatically and processed in accordance with the terms and rights of the content.

11. Limitation of blockchain in DRM

The use of encryption in DRM is secure as long as the key is secure. The encryption keys should be carefully protected and secured and still be accessible to the required users. After the data is decrypted the encryption process can no longer secure the data and prevent its distribution.

DRM systems may cause restrictions on sharing data, so to make DRM systems shareable, the traders or data providers have a little information about the protective schemes that may bring the system into a state of vulnerability.

A leak in the system may compromise the entire system of DRM content. The improper use of the DRM tools such as collecting and reporting data for personal gain may cause serious implications.

The fingerprint scanning process requires the existence of a trustworthy third party responsible for producing fingerprints as well as tracking copyright infringement. In a few other schemes that have avoided the use of such a group, it is computerized and communication levels are very high due to the use of technology like encryption. The use of blockchain in DRM makes the system vulnerable to more attacks, and some nodes may reverse the processes to process double spending and prevent others from accessing the system.

The privacy and security of users' resources may be at risk. The IP addresses can be marked and traced, and other information can be detected to get the real identities of the users through connected nodes.

There are also scalability problems in blockchain. Unauthorized blockchains make data available, allowing its public access to all stakeholders. This could jeopardize data privacy. In addition, in the event that sensitive or confidential data is loaded by mistake into the public blockchain, there is no way to repair the damage.

Though the concept of blockchain is revolutionary and to be used widely in the industry in the future, integrating blockchain with almost all management-based systems will not be fully efficient to perform. The concept of blockchain is often misunderstood as storing and managing data, which is very similar to what a database management system is used for. It is

very important to understand that if blockchain is used for the sake of maintaining records of data, it is very much not required and is unnecessary to perform.

The major advantage of using blockchain over a database management system is that it helps in distributing the data instead of centralizing it. Another better feature of blockchain is that it is designed in such a way that it provides a better way to prevent the problem of double spending. However, in case of database systems, similar functions can be performed, but the users should have full trust in the administrator of the database.

Considering the DRMS, the use of blockchain is still less advantageous. This is because a blockchain is useful when the record should not be copied; however, in this case, the record is copyright information and not the actual content. The problem is content can still be copied, whereas it is the copyright information that is actually protected. The only way to protect data is to encode data in the blockchain, but the content can inevitably escape the blockchain.

12. Conclusion and future research

One of the most important ways to curb the issue of piracy on the internet is DRM, which is a process of combining different secure technologies to provide protection to the owners and secure their ownership rights. It provides certain rules and guidelines across all forms of digitized content, mainly depicting the process of distribution and sharing. The data are converted into different formats and are flagged in such a way that they are available only to trustworthy devices and users. This prevents unauthorized users to access content and illegally share precious digital content. These guidelines can also restrict useful processes such as backup maintenance, accessing the content of different devices by the same owner, and so on.

There will always remain a demand for digital content, and people will continue to access and use it. New digital media content like songs, movies, videos, compositions, news articles, and photos are sources of entertainment for the mass users. As the content in the digital media increases, it becomes much more difficult to handle the millions of transactions that take place every second for the consumption of such content. The illegal copying of data causes the increase in the transaction costs and deliberately leads to an inefficient system of distribution.

To make DRM safe and efficient, it is very important to integrate the same with blockchain technology. This new process of using DRM with

blockchain is also being introduced by many companies to protect copyrighted content. As bad as videos of unlawful distribution of rights in the blockchain and the loss of record-keeping rights, some new concepts were introduced using the DRM blockchain.

The blockchain technology comprises cryptographic hashing algorithms, consensus protocol, and hash chains. The blockchain can be used in various aspects of DRMS. It can be used to provide irreversible traceability for digital contents; it can be used to provide monitoring of the content being accessed. It provides the capability for DRM in network media. The main characteristics of blockchain are data verifiability and being tamper-proof and decentralized in nature. There are various researches being carried out in this field to improve the governance and technical aspect of blockchain. A well-known research is the Interplanetary File System. From the application of blockchain in the industry, we can be sure that blockchain is not just a buzzword: it is going to impact every aspect of technology in the long run.

As a whole, the thought of using blockchain with DRM may still have loopholes due to the poor understanding of a copyright, blockchain, or both. This is because of the faulty understanding of blockchain solving the double spend problem, which causes the issue of protecting content and not the copyright.

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